🔥 FireBot F95 – Complete Step-by-Step Build & Working Procedure

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Step 1: Understanding the System

Before building, the team reviews the complete working of FireBot F95.

- FireBot continuously scans its surroundings using flame and infrared sensors.
- When fire is detected, it identifies the direction and center (centroid).
- The robot moves toward the fire while avoiding obstacles.
- A water jet sprinkler is activated to extinguish the fire.
- Alerts are generated and monitoring data is displayed.

This ensures all team members share a common understanding.

Step 2: Finalizing Components & Layout

The team finalizes all components and their placement:

- Chassis at the base
- Motors and wheels on sides
- Battery and controller at center
- Flame sensors placed around the robot for 360° detection
- Ultrasonic sensor at the front
- Water tank and pump on the upper platform
- Servo-mounted nozzle at the front/top

A rough hand-drawn layout is sufficient.

Step 3: Assembling the Base Structure

- Fix motors firmly to the chassis.
- Attach wheels and ensure free rotation.
- Mount battery and motor driver securely.
- Verify that the robot moves straight and turns correctly.

At this stage, the robot should **move smoothly** when powered.

Step 4: Building the Upper Body

- Construct a lightweight box using cardboard or thermocol.
- Provide openings for sensors, nozzle, and wiring.

- Line inner surfaces with aluminium foil for heat protection.
- Secure the water tank and pump to avoid vibration.

The body should be **stable, lightweight, and accessible**.

Step 5: Sensor Installation & Wiring

- Install flame sensors around the robot for full coverage.
- Fix ultrasonic sensor at front for obstacle detection.
- Mount infrared / temperature sensor for heat intensity sensing.
- Add buzzer and indicator LEDs.
- Complete wiring carefully and label connections.

Ensure no loose wires or short circuits.

Step 6: Controller Programming & Logic Development

- Upload base motor control code to Arduino.
- Add logic for:
    - Fire detection
    - Direction determination
    - Obstacle avoidance
- Implement centroid-based targeting so the nozzle aligns with fire center.
- Program pump activation and servo control.
- Add manual override using Bluetooth (optional).

Test each logic section independently.

Step 7: Fire Extinguishing Mechanism Setup

- Connect pump through relay module.
- Align nozzle using servo motor.
- Adjust water flow rate to avoid splashing.
- Ensure water reaches flame safely and accurately.

Test with **small controlled flames only**.

Step 8: Alerts & Monitoring Integration

- Configure buzzer to activate on fire detection.

- Display temperature level and fire intensity.
- Monitor water percentage and alert when low.
- Enable live camera feed and night vision (if included).
- Configure mobile alerts if supported.

This improves awareness and safety.

Step 9: Testing, Debugging & Optimization

- Test movement without fire.
- Test fire detection sensitivity.
- Test obstacle avoidance accuracy.
- Test extinguishing performance.
- Adjust delays, speed, and sensor thresholds.

Repeat tests until performance is reliable.

Step 10: Final Demonstration Preparation

- Clean wiring and secure all components.
- Prepare PPT and explanation flow.
- Practice live demo sequence.
- Assign speaking points for presentation.
- Ensure safety measures during demo.

The robot should be **stable, predictable, and safe**.